

Below are the job descriptions and details about a new job post for process engineer. Help. Me to prepare with questionnaires and answers. Starts with "why I think my experience and competency make my self believe that I am a good fit for the job." Then go bullet points requirements. To help I am attaching my CV to access myself by you. feel free to ask any information about me or job you may additionally need.

[Job purpose and primary objectives:

This role plays a key part in the assurance of chemical engineering principles within an R&D business and will provide oversight and verification of the standardised engineering delivery and detailed engineering solutions.

The role will also provide operational line technical query support to the operating teams across our operating sites.

Success is developing and assuring our inherently safer design, safe systems of work, whilst facing the challenges of a research, development and technology driven business.

Key responsibilities include:

- * Provide day-to-day chemical and process engineering support to research teams, operations, and infrastructure assets, delivering integrated engineering solutions across the wider engineering team.
- * Support resolution of operational issues by monitoring equipment reliability, troubleshooting performance, and proactively identifying improvement opportunities.
- * Lead process design activities for projects, ensuring compliance with bp Engineering Technical Practices (ETPs), HSSE requirements, and inherently safer design principles.
- * Provide oversight, assurance, and technical challenge of engineering contractor process deliverables.
- * Act as a coordination focal point between contractors, bp SMEs, and operations teams to ensure effective technical integration.
- * Support modification, upgrade, and optimisation of assets and equipment owned or operated by site teams.
- * Lead and support Management of Change (MOC) activities using approved assurance processes, ensuring deviations are formally assessed, authorised, and documented.
- * Validate refinery or facility operating data used in process design and ensure designs reflect future operating scenarios and system transitions.
- * Own and maintain Piping & Instrumentation Diagrams (P&IDs) and Major Accident Frequency (MAF) registers, ensuring full compliance through change management.
- * Lead and participate in hazard identification and risk assessment activities, including HAZID, HAZOP, PHSSER, and LOPA.
- * Ensure safe operating limits, pressure relief systems, and safeguards meet bp engineering and process safety requirements.

- * Provide process safety input to MOCs and support Process Safety Reviews (PSRs).
- * Support identification, definition, and delivery of capital investment programmes for infrastructure assets, including commissioning and initial operation of new equipment.
- * Support the Senior Chemical Engineer in maintaining site process safety and engineering integrity, contributing to development and continuous improvement of site engineering systems, processes, and procedures.
- * Support FEED and pre-FEED scoping, particularly for concept evaluation and innovation projects.
- * Select system components to ensure material suitability and chemical compatibility for defined operating conditions.
- * Provide assurance of pressure vessel design, inspection, testing regimes, code compliance, and technical analysis.
- * Deliver risk-based assurance for safety-critical equipment, supporting development and implementation of planned preventative maintenance strategies.
- * Provide technical assurance of third-party design proposals.
- * Maintain awareness of emerging technologies and advise on their appropriate application.
- * Deliver technical training and mentoring to engineers and operations personnel, supporting capability development and continuous improvement.
- * Participate in Root Cause and Failure Analysis (RCFA) to address reliability and safety issues.
- * Assure ongoing technical integrity of mechanical and process systems through active development and review of P&IDs and Process Flow Diagrams (PFDs).]

Here is a comprehensive interview preparation guide tailored specifically to your background and the target process engineer role.

This guide begins with your introductory pitch, followed by a detailed breakdown of core requirement areas. Each section connects the job's expectations directly to your 12+ years of experience at Shell, Süd-Chemie, and SABIC.

Opening Statement

"Why I think my experience and competency make myself believe that I am a good fit for the job:"

"I believe I am an excellent fit for this role because it sits at the exact intersection of my core strengths: enforcing chemical engineering principles, ensuring rigorous process safety, and managing technical delivery within an R&D and technology-driven environment. Over my 12-year career at major energy and chemical companies like Shell and SABIC, I have specialized in laboratory operations, lab-scale equipment design, and infrastructure planning.

I have a proven track record of taking complex R&D challenges and translating them into safe, compliant, and practical engineering solutions. I am deeply familiar with managing process safety tools like HAZOP, LOPA, and Management of Change (MOC) to embed inherently safer design principles into experimental setups. Furthermore, my background as a technical focal point—collaborating directly with third-party vendors, engineering contractors, and cross-functional research teams—aligns perfectly with your requirement for an engineer who can provide technical challenge, oversight, and seamless integration. I bring both the theoretical engineering discipline and the hands-on operational agility required to safeguard and optimize assets in a dynamic research environment."

Requirement-Based Questionnaires & Answers

1. Day-to-Day Engineering Support & Operational Troubleshooting

- **Requirement Focus:** Providing chemical/process engineering support, monitoring equipment reliability, troubleshooting performance, and modifying/upgrading assets.
- **Interview Question:** *"Can you describe a time when you had to troubleshoot a performance issue on a lab-scale unit or experimental skid? What was your approach?"*
- **Target Answer:**

"In R&D, troubleshooting requires balancing immediate operational uptime with long-term engineering integrity. Throughout my career, I have overseen day-to-day process engineering support for high-throughput experimentation (HTE) and test skids. For instance, when a system faces a performance bottleneck or structural deviation, my approach relies on systematic data-driven performance analysis.

At Shell and SABIC, I frequently evaluated equipment performance limits and commissioned fit-for-purpose test skids. If a skid experiences instability, I first isolate the operational variables using data protocols, check chemical compatibility for the defined operating conditions, and collaborate with the site teams to implement a safe modification. I don't just patch the issue; I ensure the modification is captured in our engineering drawings and safety limits, preventing recurrence while maintaining experimental throughput."

2. Process Design, P&IDs, and Asset Documentation

- **Requirement Focus:** Leading process design, maintaining P&IDs and PFDs, selecting system components, and ensuring code compliance.
- **Interview Question:** *"How do you ensure accuracy and compliance when designing process steps or updating critical documentation like P&IDs?"*
- **Target Answer:**

"Maintaining accurate documentation is fundamental to technical integrity. Throughout my design and commissioning experience, I have utilized Aspen modeling to simulate new and existing processes, ensuring that our fluid dynamics and thermodynamic assumptions are sound before physical execution.

When developing or updating P&IDs and PFDs, I meticulously verify component selection to guarantee material suitability and chemical compatibility under the unit's maximum operating pressures and temperatures. For example, during my time as a Lab Automation Lead at SABIC, translating manual skills into automated 24/7 systems required an absolute, line-by-line review of P&IDs to position automated valves, interlocks, and sensor taps correctly. I treat asset documentation as a living, safety-critical truth that must precisely reflect physical reality."

3. Hazard Identification, Risk Assessment & Process Safety

- **Requirement Focus:** Leading/participating in HAZID, HAZOP, LOPA; determining safe operating limits, pressure relief systems (PSVs), and safeguards.
- **Interview Question:** *"This role requires strong oversight of process safety and risk assessment. What is your experience in leading hazard reviews, and how do you define safe operating limits?"*
- **Target Answer:**

"Process safety is non-negotiable, especially in R&D where feedstocks, catalysts, and operating conditions change frequently. I have extensive hands-on experience utilizing risk assessment methodologies, including HAZOP, LOPA, What-If analysis, FMEA, and Bow-Tie analysis.

When executing DOE-based experimental campaigns at Shell, I am responsible for performing comprehensive risk assessments for each study. This involves establishing hard safe operating limits, sizing and verifying safety-critical systems like PSVs, vent/relief systems, and setting up automated emergency interlocks. By structurally analyzing the hazards beforehand, we ensure that even when pushing the boundaries of research, the physical infrastructure operates within an inherently safer design envelope."

4. Management of Change (MOC) & Technical Assurance

- **Requirement Focus:** Leading and supporting MOC activities; providing oversight, assurance, and technical challenge to contractors and third-party designs.
- **Interview Question:** *"How do you handle a situation where an operations team or a third-party contractor proposes a design change that you believe compromises technical integrity or safety standards?"*
- **Target Answer:**

"Technical assurance requires the confidence to constructively challenge proposals when safety or engineering integrity is at stake. I am highly experienced in leading Management of Change (MOC) activities for lab-scale systems and experimental setups. My approach is always rooted in formal, systematic verification.

If a contractor or internal team proposes a deviation, I utilize our approved assurance processes to ensure the deviation is formally assessed, risk-analyzed, and documented. I leverage engineering calculations and process modeling to back up my technical challenge. At Shell, conducting capability assessments of laboratory setups taught me how to evaluate equipment readiness impartially. By acting as the technical bridge between stakeholders and safety standards, I ensure that we accommodate innovation without bypassing rigorous engineering governance."

5. Capability Development, Digitization & Culture

- **Requirement Focus:** Delivering technical training and mentoring; maintaining awareness of emerging technologies; collaborating with cross-functional teams.
- **Interview Question:** *"The job description highlights mentoring and capability development. How do you upskill an operations or research team to adopt better practices or technologies?"*
- **Target Answer:**

"Capability development is a core component of my professional identity. Throughout my career, I haven't just delivered engineering solutions; I have intentionally strengthened the process engineering and safety competencies of the teams around me.

For example, at Shell, I serve as a 'DIY Coach' and digital transformation resource for the Process Development team. I champion digital literacy by training and mentoring colleagues on low-code tools and automation workflows, empowering them to independently resolve recurring operational pain points. Whether I am training technicians on safe operating limits of a new test skid or mentoring junior engineers on risk assessment documentation, I focus on building an open culture of continuous learning and strict technical discipline."

6. Capital Projects, Commissioning & Vendor Management

- **Requirement Focus:** Supporting capital investment programmes, FEED/pre-FEED scoping, commissioning new equipment, and root cause analysis (RCFA).
- **Interview Question:** "Can you tell us about your experience with equipment commissioning and working with external vendors during the early phases of a project?"
- **Target Answer:**

"I have extensive experience spanning the full lifecycle of experimental systems—from conceptual scoping to full commissioning. I have successfully commissioned multiple fit-for-purpose test skids and automated high-throughput units.

During the pre-FEED and design phases, I serve as the technical focal point, collaborating directly with automation and fabrication vendors to translate user requirements into detailed engineering specifications. For instance, at SABIC, transforming manual skids to 24/7 automated systems meant working hand-in-hand with external automation vendors to execute major technical upgrades. When issues arise post-commissioning, I utilize Root Cause and Failure Analysis (RCFA) to pinpoint reliability defects, ensuring that the final asset delivery is robust, safe, and fully integrated with our site infrastructure."

💡 Pro Tips for Your Interview:

1. **The Job Context:** The job description mentions compliance with *bp Engineering Technical Practices (ETPs)* and *PHSSER* (Project HSSE Review). Frame your experience with Shell's or SABIC's internal engineering standards as a direct parallel. Show them that you are already used to working under strict, world-class enterprise standards.
2. **Emphasize R&D Agility:** Emphasize that you understand *R&D process engineering*. Traditional plant engineers struggle with the rapid pace of changes in a lab or pilot plant. Your 12+ years of handling fast-changing experimental setups safely is a massive competitive advantage.

Would you like me to generate specific technical scenarios based on a particular type of unit operation, or drill deeper into any specific process safety methodology?

What is inherently safe design principle

The **Inherently Safer Design (ISD)** principle is a foundational approach in chemical and process engineering that focuses on **eliminating or reducing hazards at the source**, rather than relying on add-on protective equipment or human intervention to manage those hazards.

While conventional safety relies on *active* systems (like relief valves, interlocks, or alarms) and *passive* barriers (like blast walls), ISD aims to design the process so that it is *permanently and fundamentally incapable* of causing a major accident.

The Four Core Strategies of Inherently Safer Design

ISD is traditionally broken down into four distinct design strategies:

1. Minimize (Intensification)

- **Concept:** Reduce the quantity of hazardous materials present in the system at any given time. If you have less of a dangerous substance, a total loss of containment has far less severe consequences.
- **Examples:**
 - Designing a continuous reactor with a small volume instead of using a large batch reactor.
 - Storing smaller quantities of toxic gases in the lab and relying on just-in-time generation or delivery.

2. Substitute (Substitution)

- **Concept:** Replace a hazardous material or process chemistry with a less hazardous alternative.
- **Examples:**
 - Substituting a flammable or toxic solvent (like benzene) with a non-flammable, non-toxic alternative (like water or supercritical CO₂) in an extraction process.
 - Replacing a highly hazardous chemical intermediate with a more benign reactant, even if it requires a different catalytic pathway.

3. Moderate (Attenuation / Limitation of Effects)

- **Concept:** Use hazardous materials under less severe physical conditions (lower temperatures, lower pressures, less concentrated forms, or safer physical states).
- **Examples:**
 - Operating a hydroprocessing unit at a lower pressure by utilizing a more active catalyst.
 - Storing ammonia as a refrigerated liquid at atmospheric pressure rather than as a pressurized gas at ambient temperature.
 - Diluting a concentrated acid or hazard to a concentration that reduces its reactivity or vapor pressure.

4. Simplify (Simplification / Error Tolerance)

- **Concept:** Design the facility and equipment to be less complex, making it less prone to human error and less reliant on complicated control systems.
- **Examples:**
 - Designing a piping system and vessel to withstand the maximum possible pressure a pump can deliver, thereby eliminating the need for a complex pressure relief system.
 - Layout planning that ensures clear separation of incompatible materials, or using uniquely keyed fittings so operators cannot physically cross-connect incompatible chemical or gas lines.

Why ISD Matters in Process Engineering and R&D

Implementing ISD principles early in the design phase (such as during pre-FEED, FEED, or initial experimental setup) offers profound operational advantages:

- **Robustness:** Add-on safety systems (like safety valves or sensors) can fail, require regular testing, and demand constant maintenance. An inherently safer design cannot "fail" because the hazard simply isn't there.
- **Cost-Effectiveness:** While changing a process chemistry might seem expensive upfront, it significantly lowers capital and operational costs over time by reducing the need for expensive safety-critical instruments, mitigation hardware, and specialized insurance.
- **Regulatory Compliance:** Regulatory frameworks and corporate standards heavily prioritize inherently safer design during formal risk assessments like HAZOP, HAZID, and LOPA. Demonstrating that a hazard was engineered out of a system is always preferred over proving that an interlock can control it.

I want to design a safety relief valve. How dispersion model is linked to that?

When you design a **Safety Relief Valve (SRV)** or **Pressure Relief Valve (PRV)**, your primary focus is inside the process boundary: calculating the relief load, sizing the valve orifice, and ensuring the vessel doesn't overpressurize.

However, once the SRV activates, the hazard doesn't disappear—it is simply transferred elsewhere. **Dispersion modeling** is the critical bridge that links the *discharge* of the relief valve to the atmosphere, ensuring that the released chemical cloud does not pose a flammable, toxic, or asphyxiation hazard to personnel, equipment, or the surrounding community.

Here is exactly how the design of an SRV is dynamically linked to a dispersion model:

1. Input Data: The SRV Defines the Source Term

A dispersion model cannot run without a "source term"—the physical parameters of the release. The SRV design directly dictates these inputs:

- **Relief Mass Flow Rate (W):** The calculated worst-case relief load (e.g., from a runaway reaction, fire case, or control valve failure) determines the total mass of chemical entering the atmosphere per second.
- **Discharge Velocity (u):** Because SRVs typically vent at high pressures, the gas often leaves the tailpipe at sonic velocity ($Mach = 1$). This high velocity causes **choked flow**, which creates a high-momentum jet that significantly affects initial turbulent mixing (jet dispersion) in the model.
- **Temperature and Phase:** The fluid state (gas, liquid, or two-phase flashing flow) determined during SRV sizing dictates whether the plume will be buoyant (rise), neutrally buoyant, or dense (sink to the ground, like cold CO₂ or heavier-than-air hydrocarbons).

2. Setting the Physical Stack Height and Location

The dispersion model is used to verify or adjust the physical installation of the SRV tailpipe.

- If the SRV vents directly to the atmosphere (common for non-toxic, non-flammable gases like steam or air, and sometimes light hydrocarbons), the dispersion model calculates how the gas cloud dilutes as it travels downwind.
- If the model shows that toxic gas concentrations or flammable gas concentrations exceed safe limits at ground level or at nearby working platforms, **the engineer must alter the SRV tailpipe design** by raising the stack height, redirecting it away from air intakes, or routing it to a flare/scrubber system.

3. Determining the Safe Envelope (Lower Flammable Limit & Toxicity)

The dispersion model maps out a 3D boundary profile of the released gas based on specific target thresholds:

- **For Flammable Gases:** The model tracks the distance to the **Lower Flammable Limit (LFL)**—often aiming for a conservative envelope of **50% or 20% of the LFL**. The SRV discharge path must be designed so that this flammable envelope never comes into contact with ignition sources (like static electricity, hot surfaces, or non-explosion-proof electrical kit).
- **For Toxic Gases:** The model maps out exposure boundaries using standards like ERPG (Emergency Response Planning Guidelines) or AEGL (Acute Exposure Guideline Levels). The SRV must be positioned so that the toxic plume overflies populated areas safely above lethal or harmful concentrations.

4. Assessing Radiation and Noise (The Flare/Vent Link)

If the SRV vents a flammable gas that is intended to be ignited at a vent tip (or routed to a flare stack), the dispersion model is linked with a **thermal radiation model**.

- The SRV's maximum discharge rate determines the size of the flame during an emergency event.
- The model calculates the radiation heat flux (kW/m^2) to ensure that personnel on the ground have enough time to escape and that nearby structural steel does not weaken from the intense heat.

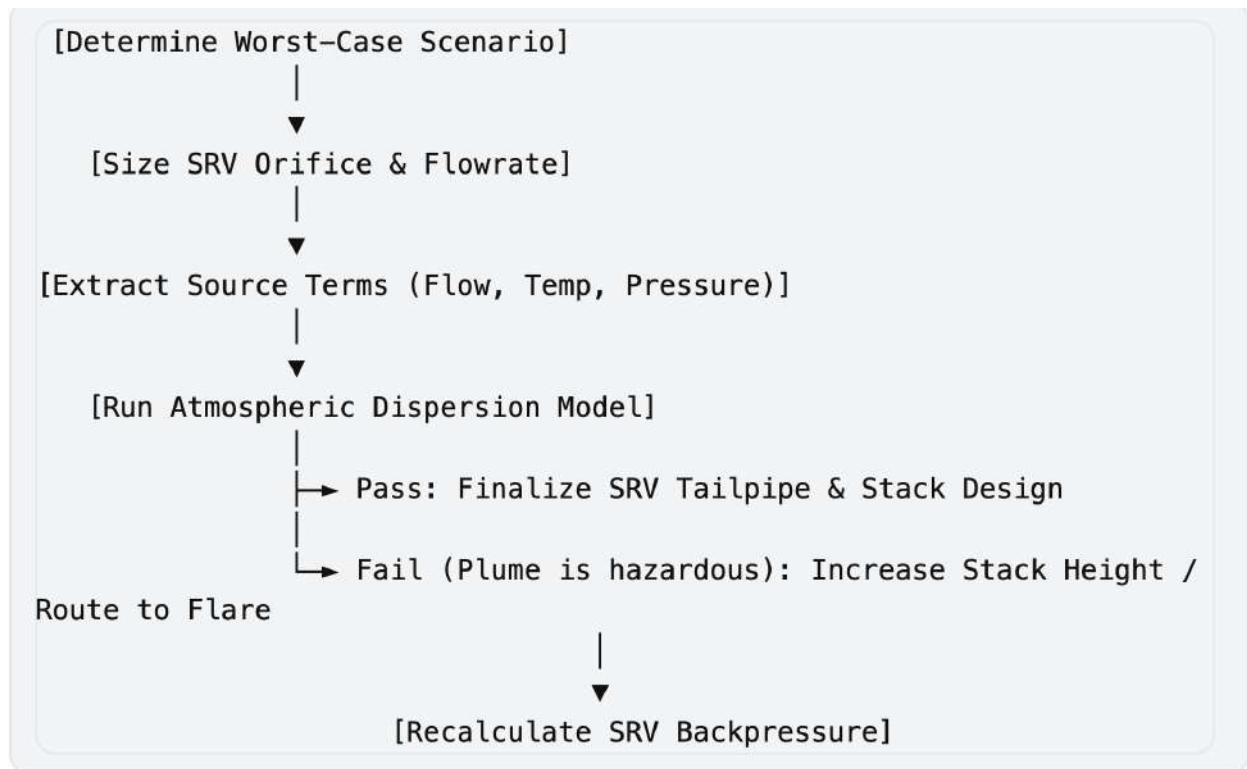
5. Backpressure and Tailpipe Sizing

There is a critical engineering feedback loop between the dispersion model and the mechanical layout of the relief system:

1. To optimize dispersion (get better mixing with air), you might want a high-velocity vertical discharge.
2. However, a restrictive tailpipe or a taller stack increases the **built-up backpressure** at the SRV outlet.
3. If the built-up backpressure exceeds the allowable limit of the valve (typically **10% of the set pressure for a conventional SRV**, or **30–50% for a balanced bellows SRV**), the valve will chatter, lose capacity, or fail to open fully.

The Design Feedback Loop: If the dispersion model says your stack must be 5 meters taller to protect workers, you must recalculate the relief piping pressure drop to ensure the added pipe length doesn't create unacceptable backpressure on your SRV.

Summary of the Engineering Workflow



By linking the two, you ensure that your design safely mitigates the overpressure hazard *inside* the vessel without accidentally creating an explosion or toxic exposure hazard *outside* the vessel.

I am operating a reactor at 200 degree C at 30 barg pressure. PRV is sized and set at 35 barg. In case of over pressure scenario how I can use dispersion model to finalised my PRV position? The gas composition in case-1 is 97% CO₂, 3% CO and case -2 is 97% CO₂ and 3% H₂

To finalize your PRV position for these two specific cases, you need to look at how the different physical and chemical properties of Carbon Monoxide (CO) and Hydrogen (H₂) alter the behavior of the released plume.

Even though both cases are heavily diluted with Carbon Dioxide (CO₂), their safety risks are entirely different: **Case 1 is a toxic/asphyxiation hazard**, while **Case 2 is a flammability hazard**.

Here is step-by-step how you use a dispersion model to determine the final, safe positioning of your PRV discharge tailpipe.

Step 1: Extract the "Source Terms" for the Model

Before running your software (such as PHAST or ALOHA), your PRV sizing calculations at the **35 barg set pressure** must provide the following inputs:

- **Total Mass Flow Rate (W):** The maximum rated relieving capacity of your selected PRV orifice at 35 barg (plus allowable overpressure).
- **Release Temperature:** Note that as the gas expands across the PRV from 35 barg down to atmospheric pressure, it will undergo significant **Joule-Thomson cooling**. Your model needs this expanded temperature because cold CO₂ becomes highly dense and sinks.
- **Velocity:** The flow at the tailpipe exit will be sonic (Mach = 1). This creates a high-momentum jet that assists initial air entrainment.

Step 2: Evaluate Case 1 (97% CO₂, 3% CO) – The Toxicity Pathway

In this scenario, your primary constraint is preventing lethal toxic exposure (CO) and asphyxiation (CO₂) at ground level or near manned platforms.

Plume Behavior:

- **Gas Density:** Carbon Dioxide has a molecular weight (MW) of 44 g/mol, and Carbon Monoxide is 28 g/mol. The average mixture MW is roughly **43.5 g/mol**.
- Because air is ~29 g/mol, this mixture is **significantly heavier than air**. Combined with Joule-Thomson cooling, this plume will be a **dense gas**. Instead of rising, it will sink toward the ground and pool in low-lying areas or pits.

Using the Dispersion Model to Position the PRV:

1. **Set the Target Threshold:** Input the toxic threshold limits for CO (such as the AEGL-2 or ERPG-2 for a 10-minute or 60-minute exposure) and the asphyxiation limits for CO₂ (typically 5% or 50,000 ppm).
2. **Map the 3D Cloud:** Run the model under worst-case meteorological conditions (typically low wind speed and stable atmosphere, like F1.5 or D5).
3. **Adjust PRV Position:** * Look at the ground-level concentration footprint. If the toxic boundary touches any walkways, building air intakes, or open workstations, **you must extend the PRV tailpipe vertically upward (stack height)**.
 - Because it is a heavy, toxic gas, venting this directly to the atmosphere inside a confined or semi-confined area is highly dangerous. The model will help you determine the minimum stack height required so that mechanical air jet mixing dilutes the CO below hazardous thresholds *before* the heavy plume falls back down to breathing zones.

Step 3: Evaluate Case 2 (97% CO₂, 3% H₂) – The Flammability Pathway

In this scenario, your primary constraint is preventing a flash fire or explosion if the released gas encounters an ignition source.

Plume Behavior:

- **Gas Density:** Hydrogen has a MW of 2 g/mol. Even though H₂ is incredibly light, it only makes up 3% of the mixture. The average mixture MW is roughly **42.7 g/mol**.
- Because 42.7 g/mol is still much heavier than air, **this mixture will also behave as a dense gas initially**, heavily dominated by the CO₂ carrier. It will not instantly float away like pure hydrogen would.

Using the Dispersion Model to Position the PRV:

1. **Set the Target Threshold:** Input the Lower Flammable Limit (LFL) of the mixture. While pure H₂ has a wide LFL (4%), it is heavily diluted here. The model will calculate the effective LFL of this specific hydrocarbon/inert blend. Usually, engineers map the **50% LFL** or **20% LFL** boundary as a safety buffer.
2. **Map the Flammable Envelope:** The model will output a 3D "flammable cloud" contour representing the zone where the gas can ignite.
3. **Adjust PRV Position:**
 - **X-Y-Z Clearance:** The PRV discharge tip must be physically positioned such that this 3D flammable envelope **never overlaps with any potential ignition sources** (e.g., non-explosion-proof electrical boxes, hot reactor walls at 200°C, or maintenance walkways).
 - **Orientation:** Ensure the PRV vents vertically straight up. Orienting it horizontally or downward will drive the flammable envelope directly into infrastructure.

Step 4: The Mechanical Feedback Loop (Backpressure Check)

Once your dispersion model gives you the safe location—for example, telling you: *"The tailpipe must vent 6 meters above the highest deck to keep the Case 1 toxic cloud away from operators"*—you must perform a final engineering validation:

$$P_{\text{built-up}} = \Delta P_{\text{tailpipe}} \leq P_{\text{allowable}}$$

- A 6-meter vertical pipe run adds significant frictional resistance.
- Calculate the **built-up backpressure** caused by this extended tailpipe during a full-flow relief event.
- If you are using a **conventional PRV**, this backpressure must not exceed **10% of your 35 barg set pressure (3.5 bar)**. If it exceeds 10%, the valve will chatter and fail.
- If the backpressure is too high, you must either **increase the diameter of the tailpipe** to lower the pressure drop, or upgrade your PRV design to a **balanced bellows type** (which can tolerate 30% to 50% backpressure).

Summary Action Plan

To finalize the position, run both cases in your dispersion software and find the **envelope maximum**. Position your PRV tip so that it clears the worst-case boundary of both scenarios: the longest toxic tail from Case 1 and the widest flammable bubble from Case 2.

How high momentum jet assists air entrainment ?

A high-momentum jet assists air entrainment primarily through **turbulent shear stress and the creation of low-pressure zones** at the boundary of the jet. This process is governed by fluid mechanics principles, specifically fluid shear, vortex generation, and the Bernoulli principle.

Here is a step-by-step breakdown of how a high-velocity, high-momentum jet (such as a gas escaping a PRV tailpipe at sonic velocity) pulls surrounding ambient air into its stream:

1. The Shear Layer and Velocity Gradient

When a high-velocity jet exits a nozzle or pipe into stationary ambient air, a sharp **velocity gradient** is formed at the boundary where the fast-moving gas meets the still air. This boundary region is known as the **shear layer**.

Shear Stress (τ) $\propto \mu \frac{dv}{dy}$

Because the velocity difference (dv) across a small distance (dy) is extremely high in a high-momentum jet, massive shear forces are generated at this interface.

2. Kelvin-Helmholtz Instability and Vortex Generation

The intense shear forces cause the boundary layer to become highly unstable, a phenomenon known as **Kelvin-Helmholtz instability**.

- This instability causes the smooth interface to ripple and break apart into microscopic turbulent eddies and swirling vortices.
- These vortices act like small mechanical rollers or gears at the edge of the jet, physically grabbing packets of stagnant ambient air and wrapping them into the main jet core.

3. Localized Pressure Drop (The Bernoulli Effect)

According to the Bernoulli principle, an increase in the speed of a fluid occurs simultaneously with a decrease in static pressure.

- The core of the high-velocity jet has a lower static pressure than the surrounding ambient atmosphere.
- This creates a radial **pressure gradient** (high pressure outside the jet, low pressure inside).
- Ambient air is naturally pushed down this pressure gradient, rushing inward toward the center of the jet to equalize the pressure.

4. Jet Expansion and Dilution

As the ambient air is continuously sucked into the shear layer, it gains momentum from the jet and begins traveling alongside it. This mass addition causes the jet to expand in diameter as it moves downstream.

As a result:

- **The jet widens** at a characteristic angle (typically around 20° to 24° for a turbulent free jet).
- **The peak velocity decreases** as the fixed kinetic energy of the jet is distributed over a continuously increasing mass of fluid (the original gas plus the entrained air).
- **Rapid dilution occurs**, which is highly beneficial in relief systems because it rapidly drops the concentration of flammable or toxic gases right at the exit tip.

Why High Momentum Matters for PRV Dispersion

In the context of relief valve design and dispersion modeling, a **high-momentum jet** is highly desirable because of how it alters the hazard envelope:

- **Maximized Dilution Rate:** Higher momentum means stronger turbulent mixing. The faster air is entrained, the quicker a toxic gas is diluted below its harmful threshold (e.g., CO in Case 1) or a flammable gas drops below its Lower Flammable Limit (e.g., H₂ in Case 2).
- **Overcoming Negative Buoyancy:** As noted in your reactor scenario, cold CO₂ mixtures are dense and want to sink. A high-momentum sonic release creates a vertical "momentum plume" that can force the heavy gas high into the air, buying time and distance for atmospheric dilution before gravity can pull the dense cloud back down to the breathing zone.

Coming back to the interview questions: help me to answer the question what are my day to day experience which I think help me to be a suitable for this job requirements

To answer this question effectively, you need to connect your daily routine as a Senior Associate Researcher at Shell and your past roles directly to the operational heartbeat of this Process Engineer position. The interviewers want to see that you don't just understand engineering on paper, but that you live it every day in the lab and pilot plant.

Here is a structured, high-impact way to frame your day-to-day experience, tailored specifically to bridge your CV and their job requirements:

The Strategic Framework (The "Day-in-the-Life" Pitch)

"In my current role at Shell and throughout my 12+ years in R&D environments, my daily routine has been a continuous cycle of translating research goals into safe, robust engineering realities. I don't just sit behind a desk; my day-to-day is split between the engineering desk, the risk assessment table, and hands-on operational troubleshooting.

Here is how my daily activities map directly to what you need on day one:"

Core Day-to-Day Experiences Mapped to Requirements

1. Daily Operational Support & Asset Troubleshooting

- **What they want:** Someone to provide day-to-day chemical and process support, monitor reliability, and proactively troubleshoot performance issues.

- **Your Daily Experience:**

"On any given day, I am the operational focal point for our high-throughput experimentation (HTE) and lab-scale test skids. I review equipment performance metrics, evaluate operational limits, and step in immediately when a system faces instability or a process deviation. My daily routine involves tracking down why a system isn't performing as expected, analyzing the data protocols, and working directly with technicians to optimize the equipment."

2. Living and Breathing Management of Change (MOC)

- **What they want:** Leading and supporting MOC activities, verifying deviations, and ensuring safety compliance

- **Your Daily Experience:**

"Because R&D involves continuous experimentation, design configurations change rapidly. Almost daily, I am initiating, reviewing, or closing out Management of Change (MOC) workflows. I systematically evaluate proposed modifications to our lab setups, ensuring every change is formally risk-assessed, documented, and officially authorized so that we never compromise on safety standards."

3. Executing Daily Risk Assessments (HAZOP / LOPA)

- **What they want:** Leading and participating in hazard identification (HAZID, HAZOP, LOPA) and managing safe operating limits.

- **Your Daily Experience:**

"I do not treat HAZOP or LOPA as occasional corporate checkboxes; they are part of my routine engineering workflow. When preparing new DOE-based experimental campaigns, I perform comprehensive risk assessments. My day-to-day engineering calculations involve establishing hard safe operating limits, reviewing chemical compatibility, and verifying that safeguards—like our vent/relief systems and automated interlocks—are perfectly sized for the upcoming chemistry."

4. Technical Integration & Vendor Collaboration

- **What they want:** Acting as a coordination focal point between contractors, SMEs, and operations.

- **Your Daily Experience:**

"A massive chunk of my day is dedicated to stakeholder management and technical coordination. I routinely serve as the core technical bridge when commissioning new equipment or executing automation upgrades. I am the person who takes the high-level needs of the research scientists, translates them into rigid engineering specifications for external automation or fabrication vendors, and ensures the final asset integrates smoothly with site utilities."

5. Cultivating a Culture of Capability & Mentoring

- **What they want:** Delivering technical training, mentoring personnel, and driving continuous improvement.

- **Your Daily Experience:**

"Daily, I act as a technical mentor and 'DIY Coach' within my department. Whether I am upskilling colleagues on process safety competencies, helping them understand equipment limitations, or training them on low-code tools to automate data analytics, capability development is naturally embedded in my day. I am wired to constantly find recurring operational pain points and train the team on how to engineer them out."

Wrap-up Summary for the Interviewer

"Ultimately, my daily experience has taught me exactly how to balance the agility that an R&D business demands with the uncompromised engineering discipline and safety integrity that a world-class energy company requires. I am already doing the heavy lifting of process safety, asset documentation, and stakeholder coordination every single day—which is why I can step into this role and deliver immediate value."

How to answer for a new capability development how do I plan my utilities and execute with the help of site management team ?

To answer this question effectively in your interview, you want to showcase your structured approach to engineering design (from concept to commissioning) while highlighting your strong stakeholder management skills.

In an R&D environment, site management team controls the master infrastructure (such as main headers for nitrogen, cooling water, power grids, and ventilation). Your job as the Process Engineer is to prove that you can precisely define your unit's requirements without overloading or disrupting the existing site infrastructure.

Here is a structured, step-by-step framework to deliver a winning answer, directly connecting your CV experience at Shell and SABIC to the job requirements:

The Strategic Framework (The 4-Step Process)

"When developing a new capability or installing a new experimental skid, I utilize a rigorous, four-phase execution strategy: **Quantify, Assess, Collaborate, and Commission**. This ensures that utility planning is technically sound, safety-compliant, and fully aligned with the site management team from day one."

Phase 1: Quantify the Utility Demand (The Process Engineering Basis)

First, explain how you calculate exactly what the new capability needs.

- **Mass & Energy Balances:** "My first step is to establish the design basis. I utilize process modeling tools like ASPEN or engineering calculations to determine the exact peak and continuous consumption rates for all required utilities:
 - **Electrical:** Power requirements (voltage, phase, amperage) for heating elements, pumps, and control systems.
 - **Gases & Fluids:** Flow rates (Nm³/hr) and delivery pressures (barg) for process gases (like H₂, CO₂, N₂) and cooling water or chilled loops.
 - **Ventilation/Exhaust:** Volumetric flow rates required to safely vent process relief streams or maintain local hood face velocities."
- **Chemical Compatibility:** "I review material suitability and chemical compatibility for the defined operating conditions to select the right piping materials and tie-in components."

Phase 2: Conduct a Site Capability Assessment (Identifying Gaps)

Show that you evaluate how your new unit fits into the bigger picture.

- **Infrastructure Evaluation:** "With my calculated utility load, I review the existing site infrastructure drawings and utility capacity logs. I check if the local distribution manifolds, electrical sub-panels, or exhaust headers can handle the additional load without starving adjacent experimental units."
- **Risk-Based Scoping:** "I identify if any infrastructure upgrades are necessary—such as running a new high-pressure gas line or upgrading a breaker panel—and include this in the initial pre-FEED scoping phase."

Phase 3: Collaborate with the Site Management Team (Technical Integration)

This is where you directly address how you partner with the site teams and contractors.

- **Technical Challenge & Review:** "I act as the core coordination focal point between the project team, site SMEs, and engineering contractors. I present my utility flow schemes, P&IDs, and spatial layout proposals to the site management team for technical challenge and assurance. This ensures my design complies with local site engineering procedures and corporate standards (such as bp ETPs)."
- **Formal Risk Assessment & MOC:** "Before a single pipe is modified, I lead or participate in formal Management of Change (MOC) and hazard identification activities (HAZOP/HAZID). We look specifically at the tie-in points. For example, we assess the risk of cross-contamination, check that check valves and isolation blinds are correctly positioned, and ensure that our pressure relief systems (PSVs) and interlocks interface correctly with site utility trip systems."

Phase 4: Execution & Commissioning (Safe Delivery)

Explain how you transition from design to physical execution.

- **Permit-to-Work & Scheduling:** "During the installation phase, I work closely with site operations to coordinate the construction schedule, ensuring tie-ins are done during planned shutdowns to minimize operational disruptions."
- **Commissioning & Testing:** "Once installed, I oversee the equipment commissioning phase. This includes pressure testing lines, checking for chemical compatibility leaks, verifying electrical rotations, and performing risk-based assurance on all safety-critical equipment. Finally, I hand over the updated asset documentation—including finalized P&IDs—to the site team for ongoing operational lifecycle management."

Bringing it Home with an Example from Your CV

To make this answer unforgettable, anchor it in your actual experience:

"I applied this exact methodology during my time as the Lab Automation Lead at SABIC and when commissioning test skids at Shell. When transforming manual skids into automated, unattended 24/7 systems, the utility profiles changed dramatically. By precisely calculating the new electrical and instrument air loads, running HAZOPs on the automated safety interlocks, and working hand-in-hand with site facilities to secure reliable, continuous utility headers, we successfully enabled safe, unattended operations without a single layout or safety incident."

For a new MOC or assets commissioning what and when to prefer HAZOP, DSR,LOPA,HAZID etc

When managing a **Management of Change (MOC)** or **Asset Commissioning** project, you cannot run all risk assessments at the same time. They are designed to be deployed sequentially as a project moves from a blank sheet of paper to live operation.

Selecting the right tool depends entirely on **the phase of the project** and **the complexity of the hazard**. Here is the roadmap of what to prefer and when to use it, structured to align directly with your process safety experience.

1. HAZID (Hazard Identification)

- **What it is:** A high-level, macro-scale brainstorming session used to identify major, broad hazards (e.g., environmental, geographic, catastrophic structural failures, or major chemical incompatibilities).
- **When to prefer it:** **Pre-FEED / Conceptual Design Phase.**
- **Application in MOC/Commissioning:**
 - Use HAZID at the very beginning of an MOC when you are deciding *where* to place a new asset or *which* process chemistry path to choose.
 - It answers big-picture questions: *Is the new test skid located too close to an exit? Does the chemical choice introduce a toxicity hazard that the building's HVAC cannot handle?*

2. HAZOP (Hazard and Operability Study)

- **What it is:** A highly structured, detailed, line-by-line, vessel-by-vessel assessment using specific guide words (e.g., *More Flow, Less Temperature, Reverse Pressure*) to identify process deviations that could lead to a hazard.
- **When to prefer it:** **Detailed Engineering Phase (Frozen P&IDs) / Prior to MOC Approval.**
- **Application in MOC/Commissioning:**
 - You prefer a HAZOP when your P&IDs, equipment specifications, and material selections are fully defined but *before* physical fabrication or installation begins.
 - For an MOC, if you are modifying a piping run or changing a reactor's operating envelope (like your 200°C, 30 barg scenario), you run a HAZOP to see how that change impacts upstream and downstream nodes. You must do this early enough to modify the physical design if a hazard is uncovered.

3. LOPA (Layer of Protection Analysis)

- **What it is:** A semi-quantitative risk assessment tool used to evaluate the *adequacy* of existing safeguards against a specific, high-consequence scenario identified during a HAZOP. It calculates if the frequency of an event meets corporate risk tolerance limits.
- **When to prefer it: Post-HAZOP / Detailed Design Finalization.**
- **Application in MOC/Commissioning:**
 - You do *not* run a LOPA for an entire system. You prefer a LOPA **only when a HAZOP identifies a severe consequence** (e.g., vessel rupture, toxic gas release) and you need to prove your safeguards are strong enough.
 - If the HAZOP reveals that a control valve failing open could overpressurize your reactor beyond 35 barg, you use LOPA to determine if a standard PRV is sufficient, or if you must add an independent Safety Instrumented System (SIS) loop with an automated interlock to achieve a specific Safety Integrity Level (SIL).

4. DSR / PSR (Design Safety Review / Pre-Start up Safety Review)

(Note: In many corporate frameworks like bp's PHSSER, this encompasses Design Safety Reviews and Pre-Startup Reviews)

- **What it is:** A rigorous physical and documentation checklist to verify that the asset was actually built exactly as designed, all safety actions from HAZOP/LOPA were implemented, and the asset is physically ready for chemicals.
- **When to prefer it: Late Commissioning / Just Prior to Introducing Hazardous Materials.**
- **Application in MOC/Commissioning:**
 - You prefer a PSR/DSR at the very tail-end of the commissioning phase. This is the final gateway before signing off on the MOC to introduce live process fluids.
 - It involves walking down the physical skid with the site management team to verify: *Are the PRV tags matching the documentation? Are the interlocks tested and calibrated? Are operators trained on the new safe operating limits? Are the live P&IDs signed off?*

Summary Timeline Matrix for Your Interview

To impress the panel, you can present this clean operational timeline for a new asset or major MOC:

Project Phase	Preferred Tool	Primary Objective	Your CV Connection
1. Conceptual Scoping	HAZID	Identify macro hazards, functional constraints, and environmental risks.	Pre-FEED assessment of lab-scale unit locations.
2. Detailed Design	HAZOP	Line-by-line deviation analysis using P&IDs to find operational gaps.	Executing DOE-based risk reviews for new experimental campaigns.
3. Risk Gap Verification	LOPA	Mathematically verify if safety-critical barriers (PSVs, Interlocks) provide enough protection.	Sizing relief/vent systems and specifying safety interlocks.
4. Pre-Commissioning	DSR / PSR	Final physical walkdown, loop testing verification, and paperwork sign-off.	Equipment commissioning and site management alignment.

Interview Tip: "In my day-to-day work, I ensure that we don't treat these tools as interchangeable. If you run a HAZOP too early without frozen P&IDs, you waste engineering hours because the design will change. If you run a LOPA without a HAZOP, you miss the initiating causes. I ensure we use the right tool at the right gate to maintain technical integrity efficiently."

Can help me to answer "what is process engineering means to me" in the context

To answer "**What does process engineering mean to you?**" in this specific interview context, your response needs to bridge the gap between pure academic engineering and the fast-paced, highly variable nature of an R&D/technology environment.

The interviewers are looking to see your personal philosophy. They want to know if you see process engineering just as "calculating pressure drops and sizing valves," or if you understand it as a vital strategic tool that enables innovation to happen safely and reliably.

Given your 12+ years of experience across Shell, SABIC, and Süd-Chemie, here is a compelling, high-impact way to frame your answer:

The Structure of Your Answer

1. **The Core Philosophy:** Define process engineering as the "bridge" between an idea and reality.
2. **The R&D Twist:** Explain how process engineering changes in a research/technology setting compared to a traditional large-scale manufacturing plant.
3. **The Anchor (Safety & Discipline):** Emphasize that process engineering is the guardian of safety limits and data integrity.
4. **The Personal Connection:** Bring it home using your background.

Suggested Script / Answer Blueprint

"To me, process engineering is the discipline of translating complex scientific concepts and experimental ideas into safe, scalable, and highly reliable physical realities. > In a traditional manufacturing refinery, process engineering is often about maintaining a steady state and optimizing an existing large-scale asset. But in a **dynamic R&D and technology-driven environment like yours**, process engineering means something much more agile. It means being the fundamental bridge between a scientist's research goals and the physical limitations of the infrastructure.

To me, it specifically means three things:

1. **Designing for Inherently Safer Operations:** It means looking at a blank canvas or a new experimental setup and proactively engineering the hazards *out* of the system using tools like HAZOP, LOPA, and material compatibility analysis—rather than just adding layers of alarms and hoping they work.
2. **Enabling Agility Safely:** R&D requires constant modification, new chemistries, and rapid changes. Process engineering to me isn't about saying *'No, we can't do that experiment because it's too risky.'* Instead, it's about using rigorous Management of Change (MOC) workflows and process logic to find the exact pathway to say, *'Yes, we can do this experiment, and here is how we modify the system to do it safely.'*
3. **Securing Technical Integrity:** It means ensuring that every piece of documentation—from a P&ID to a safe operating limit—is an absolute, living truth. If our documentation or our process control fails, our data integrity fails, and more importantly, our people are put at risk.

Over my 12 years at companies like Shell and SABIC, this philosophy has guided me daily. Whether I was transitioning manual rigs into automated 24/7 systems or running risk assessments for complex, high-throughput campaigns, process engineering has never been just desk-work to me. It is an active, day-to-day commitment to safeguarding our people while accelerating the technology and innovations of tomorrow."

Why This Answer Works for This Role:

- **It Speaks Their Language:** The job description mentions *"facing the challenges of a research, development, and technology-driven business."* This answer directly highlights that you understand R&D process engineering is different from routine plant engineering.
- **It Targets Key Responsibilities:** It touches upon inherently safer design, MOCs, P&IDs, and stakeholder collaboration without sounding like you're just reading off a list.
- **It Highlights Value:** It frames you not as a bottleneck or a "bureaucratic inspector," but as an *enabler* who uses engineering discipline to help research teams succeed safely.

Help me to answer why I apply this post or want to change the company

When answering **"Why are you applying for this position?"** or **"Why do you want to leave your current company?"**, your response must always remain positive, forward-looking, and focused on growth. Because you are transitioning from one major energy player (Shell) to another (bp), the interviewers will want to see that you are looking to scale up your responsibilities and impact, rather than just running away from your current job.

The job description emphasizes a critical need for someone who can provide **technical assurance, contractor oversight, and a strong challenge to detailed engineering solutions**. Your answer should highlight that while you have built a world-class foundation in R&D operations and safety workflows, you are ready to pivot that expertise into a dedicated, high-level engineering assurance role.

Here is a structured, strategic way to answer this question:

The Core Strategy

1. **Praise the Present:** Briefly acknowledge that you've had a great journey building robust competencies where you are.
2. **Pivot to the Future:** State that you are ready for a role that centers purely on rigorous engineering assurance and contractor oversight, which aligns perfectly with this post.
3. **Align with the Company:** Express excitement about bp's engineering technical practices (ETPs) and their commitment to inherently safer design.

Suggested Script / Answer Blueprint

"I have had a fantastic journey over the last several years, where I've been able to build deep, hands-on expertise in lab-scale equipment design, high-throughput operations, and complex process safety workflows like HAZOP, LOPA, and Management of Change (MOC). I am incredibly proud of the safety-critical competencies and digital transformation workflows I've delivered.

However, I am applying for this position because my career goals have evolved, and I am looking to step into a role focused entirely on **standardized engineering delivery, technical assurance, and rigorous contractor oversight**.

When I looked at your job description, it felt like the natural next step for my professional growth for three main reasons:

1. **Focus on Technical Assurance:** While my current role involves a blend of research operations and engineering, this position allows me to focus fully as a Process Engineer who provides technical challenge, verifies design proposals, and maintains the strict engineering integrity of assets. I want to be the gatekeeper for engineering standards.
2. **Alignment with bp's Engineering Culture:** I have massive respect for bp's Engineering Technical Practices (ETPs) and your structured approach to inherently safer design. Coming from a peer company in the energy sector, I speak this language fluently and want to apply my 12+ years of experience to uphold and continuously improve these exact systems.
3. **The R&D and Technology Challenge:** I thrive in dynamic, technology-driven business environments where configurations change rapidly. I know firsthand the unique challenges of keeping a research site safe without choking innovation. I am looking to change companies because this specific role offers the exact platform where my specialized background in R&D process safety can deliver immediate, high-impact value to your operating teams."

Key Takeaways for the Interview:

- **Why this works:** It completely avoids any negative talk about your current employer. Instead, it frames your move as a natural, ambitious progression toward a role with a heavier focus on *engineering governance and contractor management*.
- **Connecting to "CV-Draft2026 (2).docx":** By explicitly referencing your 12+ years of design, commissioning, and process safety experience across major organizations, you reinforce that you aren't a novice; you are a seasoned professional ready to step up as a technical focal point.

Tell me about your experience in process engineering

When answering "**Tell me about your experience in process engineering**," you want to present a structured overview that connects your 12+ years of specialized background directly to the technical assurance and asset integrity needs of the role.

Here is a high-impact, professional response structured to showcase your journey across major energy and chemical players like Shell, SABIC, and Süd-Chemie:

The Overview (The 12+ Years Framework)

"Over my 12+ year career in the energy and chemical sectors, I have developed deep, hands-on process engineering expertise specializing in laboratory operations, lab-scale equipment design, infrastructure planning, and safety-critical system management.

Because my experience sits at the intersection of process development and R&D technology, my engineering approach is defined by two main strengths: maintaining rigid technical integrity and engineering hazards out of highly dynamic, rapidly changing systems.

My core experience can be broken down into four key engineering pillars:"

The Four Pillars of Your Process Engineering Experience

1. Process Design, Simulation & Specification

- **Modeling & Scale-Up:** I regularly utilize ASPEN to model both new and existing chemical processes, scoping out workflows to enhance operational competitiveness and optimize asset functionality. At Süd-Chemie, I managed the data protocols for scaling up catalyst synthesis from lab-scale to commercial production.
- **Component Selection & Documentation:** My daily engineering calculations include verifying material suitability and chemical compatibility under extreme or varied operating conditions. I have a strong background in developing, actively reviewing, and maintaining Piping & Instrumentation Diagrams (P&IDs) and Process Flow Diagrams (PFDs) to ensure they remain absolute, living truths for the operations teams.

2. Process Safety Management (PSM) & Risk Assessment

- **Hazard Identification:** I have comprehensive experience leading and participating in rigorous risk assessment methodologies, including HAZOP, LOPA, What-If analysis, FMEA, and Bow-Tie analysis.
- **Safeguard Sizing:** When executing complex experimental campaigns at Shell, I am responsible for establishing hard safe operating limits and sizing safety-critical systems, including Pressure Safety Valves (PSVs), vent/relief systems, and automated safety interlocks. I ensure that all process modifications undergo systematic Management of Change (MOC) workflows before physical execution.

3. Equipment Commissioning & Automation Upgrades

- **Asset Commissioning:** I have successfully designed, built, and commissioned multiple fit-for-purpose test skids and high-throughput units. This includes performing initial capability assessments of laboratory setups to ensure equipment performance and safety systems meet strict infrastructure requirements.
- **Technical Integration:** During my time as the Lab Automation Lead at SABIC, I served as the technical focal point for transforming manual experimental skids into fully automated systems, collaborating directly with third-party automation vendors to enable safe, unattended 24/7 operations.

4. Continuous Improvement & Technical Mentoring

- **Root Cause Analysis:** I utilize operational data and data-driven performance analysis to troubleshoot equipment reliability issues and support performance optimization.
- **Capability Development:** Process engineering to me includes upskilling the team. At Shell, I actively drive capability development by strengthening process engineering and safety competencies across our labs, and I act as a digital coach to help colleagues automate workflows and eliminate recurring operational pain points.

Wrap-up Summary for the Interviewer

"Ultimately, my process engineering experience has taught me how to operate with the discipline of a plant engineer while maintaining the agility required by an R&D business. I understand how to provide technical challenge to third-party designs, how to interface between contractors and internal SMEs, and how to ensure that an asset is structurally robust and inherently safe from day one."

I do not have a formal educational degree in process engineering..please incorporate this aspect too

It is incredibly common in R&D and technology-driven sectors to have brilliant engineers who entered the discipline through an applied science or chemistry pathway. Your educational background—a **Gold Medalist in Applied Chemistry and Engineering** from M S University, Baroda—is actually a massive asset, not a deficit.

In chemical engineering, having a deep, molecular-level understanding of chemistry, catalysts, and kinetics combined with 12+ years of hands-on engineering execution makes you a *stronger* troubleshooter than someone who only looks at the process as a macro-system of pipes and valves.

Here is how you can seamlessly incorporate this aspect into your key interview answers, turning it into a unique selling point (USP).

1. Updated Opening Statement / Pitch

Incorporate this right into your introduction to take control of the narrative immediately.

"I believe I am an excellent fit for this role because it sits at the exact intersection of my core strengths: enforcing chemical engineering principles, ensuring rigorous process safety, and managing technical delivery within an R&D and technology-driven environment.

While my formal postgraduate degree is as a Gold Medalist in Applied Chemistry and Engineering rather than a traditional, pure chemical engineering degree, I have spent the last 12+ years operating explicitly as a hands-on process and safety engineer at world-class energy companies like Shell and SABIC.

This unique combination gives me a distinct advantage: I deeply understand the molecular chemistry, catalyst behaviors, and reaction kinetics from my academic background, and I have paired that with over a decade of proven expertise in sizing relief systems, running HAZOP/LOPA, modeling in ASPEN, and managing asset modifications. I bring both the chemical insight and the engineering discipline required to safeguard and optimize assets in a dynamic research environment."

2. Updated Answer: "Tell me about your experience in process engineering"

Frame your experience as a masterful blend of science and engineering application.

"Over my 12+ year career, my path in process engineering has been defined by a deep integration of applied science and practical engineering design. Coming from a strong postgraduate foundation in Applied Chemistry and Engineering, I built my process engineering expertise directly in the field, managing lab-scale equipment design, infrastructure planning, and safety-critical systems at Shell, SABIC, and Süd-Chemie.

Because of this background, I don't just look at a process flow diagram as lines and blocks. I understand the fluid dynamics, the thermodynamic properties, and the chemical compatibilities at a fundamental level. Over the past decade, I have fully cross-trained and proven my competencies in core process engineering pillars:

- **Process Safety:** Sizing PSVs, vent systems, and automated safety interlocks, while leading HAZOP and LOPA reviews.
- **Process Design:** Utilizing ASPEN modeling for scale-up and process optimization, and owning the accuracy of P&IDs and PFDs.
- **Asset Commissioning:** Acting as the technical focal point for transforming manual skids into fully automated 24/7 systems.

I have proven to my peer companies that a strong scientific foundation combined with rigorous field-engineering execution creates highly robust, inherently safer designs."

3. How to handle a direct question: "We see your degree is in Applied Chemistry, not Process Engineering. How will you manage?"

If they ask directly, use this confident, peer-level response.

"That is a very fair observation. My formal educational background is a Master's degree where I was a Gold Medalist in Applied Chemistry and Engineering. However, I look at process engineering as a discipline defined by application, standards, and rigorous safety management—all of which I have executed at the highest corporate levels for 12 years.

Major global operators like Shell and SABIC trusted me to become their Lab Automation Lead, to lead Management of Change (MOC) workflows, and to size safety-critical relief systems because I have mastered the engineering technical practices required in the field.

In fact, in an R&D and technology-driven business, my background is a major differentiator. When a process deviation occurs, I can troubleshoot it from the perspective of the physical engineering constraints *and* the underlying chemical reaction kinetics. I speak the language of your research scientists fluently, but I enforce the rigid engineering design practices of a process engineer—making me highly effective at technical assurance and contractor oversight."

Pro-Tip for Your Interview:

By addressing your degree upfront as **"Applied Chemistry and Engineering"** and immediately coupling it with your **12+ years of enterprise engineering experience at Shell and SABIC**, you completely neutralize any checklist objections an HR interviewer might have. You shift the focus from a piece of paper to your undeniable track record of technical delivery.

based on the job requirement mentioned above can you prepare me for the technical interview session? help me with sets of questionnaires also

To prepare you for the core technical interview session, we need to focus heavily on the specific engineering frameworks, calculations, and standards mentioned in the job description (such as bp's Engineering Technical Practices, inherently safer design, MOC, and relief systems).

Because your degree is in **Applied Chemistry and Engineering** and you possess 12+ years of hands-on experience, the technical panel will test whether you can translate chemical kinetics and process workflows into rigid, safe physical piping and instrumentation systems.

Here is a comprehensive set of technical interview questions divided by core engineering categories, complete with the technical concepts you should highlight and a structured master answer for each.

Category 1: Process Safety & Relief System Design (PSVs/MAFs)

The job description explicitly highlights owning Major Accident Frequency (MAF) registers, ensuring pressure relief systems meet engineering requirements, and establishing safe operating limits.

Question 1: "Walk us through your technical workflow when sizing a Pressure Relief Valve (PRV) for a new reactor or gas-handling skid."

- **What they are testing:** Your knowledge of code compliance, thermodynamic behavior, and mechanical constraints.
- **Technical Concepts to Mention:** API 520/521 standards, worst-case scenario definition, choked/sonic flow, and built-up backpressure limits.
- **Master Answer:**

"Sizing a relief valve always begins with defining the **worst-case credible overpressure scenario** through a structured review—whether it's external fire, control valve failure-open, thermal expansion, or a runaway chemical reaction.

Once the scenario is established, I calculate the required relief mass flow rate (W) using fluid property data and process simulation. I determine if the fluid behavior at the valve orifice will experience **choked flow** (sonic velocity, $Mach = 1$), which is highly common in high-pressure gas releases like the 35 barg reactor scenario we discussed earlier.

After calculating the required discharge area, I select the standard API orifice size (e.g., G, H, J) that exceeds this requirement. However, the design isn't finished until I check the discharge tailpipe configuration. I run calculations to ensure that the **built-up backpressure** does not exceed **10% of the set pressure** for a conventional PRV, or up to **30–50%** if I specify a balanced bellows PRV. This systematic approach ensures the valve will open fully and operate without destructive chattering, protecting both the asset and the personnel."

Question 2: "How do you manage a Major Accident Frequency (MAF) register or maintain site process safety integrity when managing changes?"

- **What they are testing:** Your understanding of risk barriers and how asset modifications impact cumulative site risk.
- **Technical Concepts to Mention:** Bow-Tie analysis, independent protection layers (IPLs), and barrier management.

- **Master Answer:**

"An MAF register isn't just a static log; it's a live reflection of our site's defensive barriers against catastrophic failure. Throughout my experience at Shell, managing risk-critical systems meant understanding that every modification introduced via an MOC can potentially degrade or alter an existing safety barrier.

In my daily workflow, when a modification is proposed, I evaluate it against our established **independent protection layers (IPLs)**. I map out the hazard using tools like **Bow-Tie analysis** to visually confirm if the change bypasses a critical prevention barrier (like an automated safety interlock) or a mitigation barrier (like a relief vent system). Maintaining integrity means ensuring that any deviation from the baseline design pressure, temperature, or chemical compatibility is rigorously analyzed, updated in the asset registers, and communicated clearly to the operations teams."

Category 2: Technical Assurance & Contractor Oversight

The job description emphasizes providing oversight, assurance, and technical challenge of engineering contractor process deliverables.

Question 3: "If an engineering contractor delivers a set of P&IDs for a facility upgrade, what is your checklist for providing technical assurance and challenge?"

- **What they are testing:** Your attention to detail, adherence to standards, and ability to act as a rigorous technical gatekeeper.
- **Technical Concepts to Mention:** Line sizing velocity limits, isolation philosophy, chemical compatibility, and spec breaks.
- **Master Answer:**

"When providing technical assurance on contractor P&IDs, I do not just look at the flow lines; I look for hidden operational and safety gaps. My engineering checklist focuses on four major areas:

1. **Piping Spec Breaks:** I verify exactly where piping classes change—for example, where a high-pressure line drops down past a control valve into a low-pressure header—ensuring that the downstream piping and vessels are structurally rated or fully protected by a relief valve.
2. **Fluid Velocity and Sizing:** I check the line velocities against engineering technical practices to prevent problems like erosion-corrosion from high-velocity liquids or excessive pressure drops in gas lines.
3. **Isolation and Maintenance Philosophy:** I challenge the design to ensure operators can safely isolate critical components (pumps, control valves, instruments) using double-block-and-bleed configurations where required, without shutting down the entire infrastructure.
4. **Material Suitability:** Drawing from my strong **Applied Chemistry and Engineering background**, I critically challenge component selection to ensure full chemical compatibility and prevent issues like hydrogen embrittlement or stress corrosion cracking under maximum operating limits."

Category 3: Process Design, Scale-Up, & Component Selection

The job description highlights FEED/pre-FEED scoping, selecting system components for material suitability, and translating operational data into designs.

Question 4: "Can you give an example of how you used process modeling (like ASPEN) and operational data to scale up a process or optimize a design?"

- **What they are testing:** Your practical engineering capabilities and ability to use simulation tools to solve real physical problems.
- **Technical Concepts to Mention:** Equation of state (e.g., Peng-Robinson), physical property validation, and data protocols.
- **Master Answer:**

"At Süd-Chemie and Shell, a major part of my process engineering responsibilities was managing data protocols to transfer chemical and catalyst workflows into robust physical scale-up designs.

When optimizing a system, I first validate the **equation of state** within ASPEN (such as using Peng-Robinson for high-pressure hydrocarbons or NRTL for polar mixtures) to ensure the simulation accurately reflects physical chemistry realities. I then import raw operating data from our existing skids—pressure, temperatures, and composition profiles—to anchor the model.

For example, during a scale-up project, the simulation might show that a simple increase in volume would cause a runaway thermal profile due to exothermic reaction kinetics. By coupling my chemistry insight with ASPEN heat transfer modeling, I can optimize the design by selecting the preheating exchanger surface area, utility flow rates, and control valve characteristics required to maintain stable, safe operating limits."

Category 4: Hazard Methodologies (HAZOP / LOPA / RCFA)

The job description demands active leadership and participation in HAZOP, LOPA, PHSSER, and Root Cause and Failure Analysis (RCFA).

Question 5: "How do you distinguish between a HAZOP and a LOPA when determining if a system is safe enough? Can you walk us through an example?"

- **What they are testing:** Your clear understanding of qualitative vs. semi-quantitative risk assessment boundaries.
- **Technical Concepts to Mention:** Cause-Consequence pairs, Initiating Event Frequency (IEF), Probability of Failure on Demand (PFD), and Risk Tolerability Criteria.
- **Master Answer:**

"I view **HAZOP** as a **qualitative tool** designed to find *what* can go wrong, while **LOPA** is a **semi-quantitative tool** used to verify *if* our existing safeguards are strong enough to prevent it.

During a HAZOP, we use guide words like '*No Flow*' or '*More Pressure*' on a specific P&ID node. If the team identifies a cause-consequence pair with severe impacts—such as a control valve sticking open leading to overpressurization and a potential catastrophic rupture of a reactor—we pause.

We then elevate that specific scenario to a **LOPA**. In the LOPA, I assign mathematical values: the initiating event frequency of a control valve failure might be 10^{-1} per year. If our target risk limit for a major release is 10^{-5} per year, we have a risk gap of 4 orders of magnitude. I then evaluate our independent protection layers (IPLs). A standard, certified PRV might give us a Probability of Failure on Demand (PFD) of 10^{-2} . That leaves a remaining gap of 10^{-2} . I would use this technical analysis to challenge the design and justify adding a high-reliability Safety Instrumented System (SIS) loop with an automated interlock to close the gap completely."

Question 6: "Tell us about a time you used Root Cause and Failure Analysis (RCFA) to resolve a recurring operational or safety issue on an asset."

- **What they are testing:** Your analytical troubleshooting style and long-term engineering mindset.
- **Technical Concepts to Mention:** 5-Whys, Fishbone diagram, physical/human/latent roots, and corrective actions.
- **Master Answer:**

"Throughout my 12+ years managing lab-scale assets and automated test skids, I have found that recurring failures are usually symptoms of deeper latent or systemic issues. When an equipment reliability issue arises, I lead an RCFA using structured methodologies like the **5-Whys** or a **Fishbone diagram**.

For instance, at SABIC, when transforming manual experimental skids into automated 24/7 units, we experienced a series of recurring premature trips on a high-pressure line. Instead of simply replacing the pressure transmitter, I led an RCFA. We traced the physical root to micro-pulsations from an upstream positive displacement pump. The human root was that the maintenance team was calibrating the transmitter dampening factors too tightly, and the latent root was a lack of standardized engineering utility design rules for dampening pulsation in small-scale automated skids.

By identifying the true root causes, we engineered a permanent solution—installing a small surge volume and updating our design procedures—which eliminated the downtime entirely and ensured ongoing technical integrity."

Technical Cheat Sheet for Your Interview Day:

Term / Acronym	What it means to bp	How you map your experience
bp ETPs	Engineering Technical Practices (bp's strict design codes).	"I am deeply familiar with translating corporate engineering standards (like Shell's internal design rules) into daily compliance workflows."
PHSSER	Project HSSE Review (Gate-based safety review for projects).	"I map this directly to my experience leading capability assessments and equipment pre-commissioning walkthroughs."
Inherent Safer Design	Eliminating hazards at the source (Minimize, Substitute, Moderate, Simplify).	"I use my Applied Chemistry and Engineering background to substitute hazardous chemicals or moderate reactor envelopes early in design."
MOC Assurance	Formal management of facility, process, or organizational deviations.	"I have spent over a decade systematically initiating, risk-assessing, and documenting changes to avoid operational drift."